Number-average molecular weight of branched polymers from SEC with viscosity detection and universal calibration

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Abstract

BACKGROUND: Number-average molecular weight, $\overline{M}_n$, is an important characteristic of synthetic polymers. One of the very few promising methods for its determination is size-exclusion chromatography (SEC) using on-line viscometric detection and assuming the validity of the universal calibration concept.

RESULTS: We have examined the applicability of this approach to the characterization of statistically branched polymers using 22 copolymers of styrene and divinylbenzene as well as 3 homopolymers of divinylbenzene with various degrees of branching. SEC with three on-line detectors, i.e. concentration, light scattering and viscosity, enables us to evaluate experimental data by various computational procedures yielding $\overline{M}_n$ and weight-average molecular weight, $\overline{M}_w$. Analysis of the results has shown that the universal calibration theorem has limited validity, apparently due to the dependence of the Flory viscosity function on the molecular shape, the molecular weight distribution and the expansion of molecules.

CONCLUSION: For complex polymers, the universal calibration, i.e. the dependence of the product of intrinsic viscosity and molecular weight, $[\eta]M$, on elution volume, can differ in values of $[\eta]M$ from those obtained for narrow molecular weight standards by 10–15%. The method studied is helpful for the determination of $\overline{M}_n$ of polymers, in particular of those with very broad molecular weight distribution, such as statistically highly branched polymers.

Keywords: size-exclusion chromatography; triple detection; viscometric detection; number-average molecular weight; universal calibration; statistical branching; polystyrene calibration

INTRODUCTION

Number-average molecular weight, $\overline{M}_n$, is an important characteristic of polymers that are non-uniform with respect to molecular weight, and, thus, of all synthetic polymers. Theoretical analyses of mechanisms of polymerization reactions usually lead to values of $\overline{M}_n$. Unfortunately, the potential of the available experimental techniques yielding $\overline{M}_n$ is rather limited. Colligative methods, such as end-group analysis, ebullioscopy, cryoscopy or vapour-pressure osmometry, are not sensitive enough for molecular weights higher than some 15 000. Membrane osmometry is operative in a broader molecular weight range, say 20 000–500 000, but at the present time its use is rarely encountered for various reasons. MALDI-TOF mass spectrometry yields good $\overline{M}_n$ data for polymers with narrow molecular weight distributions ($\overline{M}_w/\overline{M}_n < 1.3$–$1.5$) and, preferably, not very high molecular weights.[1]

$\overline{M}_n$ can be calculated from a size-exclusion-chromatography (SEC) measurement using a concentration detector, calibration for the particular polymer and apparatus and correction for band broadening. By far the most widely used concentration detector is a differential refractometer. This detection mode becomes ineffective if the refractive index increment for a given polymer–solvent system is too small, or if the refractive index increments of copolymer components differ greatly and heterogeneity in composition is expected. Using on-line concentration and light-scattering detectors, we can calculate $\overline{M}_n$. This value, however, is associated with a non-negligible error mainly for two reasons: (i) $\overline{M}_n$ is sensitive to the low molecular weight fractions, for which the signal-to-noise ratio of the light-scattering detector is small;[2] and (ii) the molecular weight of the individual slices of the chromatogram measured is the weight average, $\overline{M}_w$, whereas $\overline{M}_n$ of the slices should be used[3] in the calculation of $\overline{M}_n$ for the whole polymer. The non-uniformity of the slices may affect the results significantly in measurements of branched polymers because linear molecules elute

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(*Received 22 April 2008; revised version received 6 June 2008; accepted 25 June 2008*)
DOI: 10.1002/pi.2458

simultaneously with branched molecules having higher molecular weight.

In the everyday use of SEC, the universal calibration (UC) pattern[4] is employed widely. This is based on Flory's theory of intrinsic viscosity[5] which suggests that the product of intrinsic viscosity and molecular weight, $[\eta]M$, is the hydrodynamic volume of a macromolecule, V_H, the quantity defining the elution volume at which a given molecule is eluted from a particular SEC set. Flory expected that his equation

$$[\eta]M = \Phi\left(\overline{s^2}\right)^{3/2} \tag{1}$$

may 'hold approximately for many nonlinear molecules as well as for linear ones'. In Eqn (1), Φ is the viscosity function and $\overline{s^2}$ is the mean square radius of gyration. Flory assumed that Φ was 'independent of the characteristics of the given chain molecule beyond the requirement that its spatial form be that characteristic of a randomly coiled chain molecule'. The UC concept appears to be justified for constant Φ. A careful analysis by Bohdanecký and Kovář[6] of theories of viscosity and published experimental data, however, has shown that Φ is not a constant but a sensitive function of the molecular weight distribution of the polymer, the topology of its molecules and the expansion of the molecules in good solvents. For polymers that are non-uniform with respect to molecular weight, number-average values of molecular weight and radius of gyration should be used.[7] Most frequently, these values are obtained by light scattering. As is well known, that method yields the weight-average molecular weight but the z-average radius of gyration. To recalculate the values to number-average values, the form of the molecular weight distribution must be known. If light-scattering data for a non-uniform polymer are used to calculate the value of Φ, an average value, Φ_{av}, is obtained. The effect of molecular weight non-uniformity is rather strong. For polymers with $\overline{M}_w/\overline{M}_n = 2$, the ratio $\Phi_{av}/\Phi = 0.51$. Even for $\overline{M}_w/\overline{M}_n = 1.1$, $\Phi_{av}/\Phi = 0.87$. The ratio of the viscosity functions for branched and linear molecules with the same molecular weight, Φ_b/Φ_l, has limiting values of 2.7 for regular stars, 1.9 for irregular stars, 2.4 for randomly branched molecules and 1.0 for comb-like molecules. The expansion of the chain molecules decreases the value of Φ by as much as 20%. Although Φ is mostly treated as a constant, in reality its value can vary almost by ±100%. These facts question the universality of the UC and indicate why for some polymers the UC concept may apply whereas to some others it does not.[8,9] For statistically branched polymers, such as those characterized in the present study, the molecular weight, the degree of branching and the breadth of the molecular weight distribution increase with conversion. This renders the variation of the value of Φ to be intricate. We prefer to refrain from speculation and regard the UC concept as roughly valid, based on the experimental evidence.

In 1989 Goldwasser[10,11] put forward the attractive idea that, assuming the validity of the UC theorem in SEC, $\overline{M}_n$ of a polymer can be obtained from the signal of just a single on-line viscosity detector and the injected mass of the polymer. Viscosity of dilute polymer solutions can be measured on-line with good accuracy and the use of only one detector avoids potential error introduced by the inaccurate determination of the inter-detector volume.[12]

In spite of the importance and theoretical soundness of Goldwasser's stratagem, its limitations and reliability in experimental application have not been examined systematically so far. For polystyrene and other narrow molecular weight distribution (MWD) linear standards, the UC postulate seems to be fulfilled relatively satisfactorily.[13,14] Detailed studies by Pang and Rudin[15,16] on linear and branched NBS polyethylene standards have shown that the fraction of the branched polymer eluting at small elution volumes and containing molecules with high molecular weight and high degree of branching does not fit the UC dependence obtained for linear standards. Recently, Farmer *et al.*[17] tested the validity of the UC concept on a number of model branched polymers with regular-comb and regular-centipede architecture in good and theta solvents, and reported that the UC plot 'appears to be effective for reducing data for branched and linear polymers to a common calibration curve'.

According to published results and personal communications, plots of $\log([\eta]M)$ *versus* elution volume may yield apparently straight or moderately curved lines over three or four orders of magnitude of values of $[\eta]M$ for chemically different polymers, and polymers with various topologies and degrees of branching. Closer inspection, however, reveals that this seeming harmony is due to a large extent to the buffering capacity of the logarithmic scale. The scatter of experimental points around the dependences corresponds, typically, to a dispersion of ±15% and, not exceptionally, to multiples in the value of the product $[\eta]M$.

Within our current extensive study of distributions with respect to molecular weight, degree of branching and architecture for various types of branched polymers, it is mandatory to operate with number-average molecular weights. Therefore, we wondered to what extent the Goldwasser method for the determination of $\overline{M}_n$ can be relied on. The present paper reports our measurements, results and conclusions.

THEORY

The number-average molecular weight is defined as

$$\overline{M}_n = \frac{\sum_i n_i M_i}{\sum_i n_i} \tag{2}$$

where n_i and M_i are, respectively, the number and molecular weight of molecules of species i in the sample.

$\overline{M}_n$ can be approximately expressed by changing the summation into integration and assuming that the number of molecules in the measuring cell of a SEC detector is proportional to the polymer mass concentration divided by its molecular weight, $n_i \propto c_i/M_i$, as[10,11]

$$\overline{M}_n = \frac{\int_{-\infty}^{\infty} c(V)\mathrm{d}V}{\int_{-\infty}^{\infty} c(V)/M(V)\mathrm{d}V} \tag{3}$$

where $c(V)$ and $M(V)$ are, respectively, the concentration and molecular weight at elution volume V.

Integration of the numerator gives the injected mass, m_{inj}. The UC, $\mathcal{J}(V)$, is defined by the product

$$\mathcal{J}(V) = [\eta](V)M(V) \tag{4}$$

At infinite dilution, the intrinsic viscosity $[\eta]$ can be replaced by the ratio η_{sp}/c and the fraction in the denominator of the integral in Eqn (3) can be rewritten as

$$c(V)M(V) = \eta_{\mathrm{sp}}(V)[\eta](V)M(V) = \eta_{\mathrm{sp}}(V)\mathcal{J}(V) \tag{5}$$

Thus we obtain for $\overline{M}_n$

$$\overline{M}_n = \frac{m_{\mathrm{inj}}}{\int_{-\infty}^{\infty} \eta_{\mathrm{sp}}(V)/\mathcal{J}(V)\mathrm{d}V} \tag{6}$$

UC is usually obtained by measuring the molecular weight and intrinsic viscosity of a series of narrow MWD standard polymers covering a large range of M and, therefore, of V. Thus

$$\overline{M}_n = \frac{m_{\mathrm{inj}}}{\Delta V \sum_i \eta_{\mathrm{sp},i}/[\eta]_i M_i} \tag{7}$$

where $\eta_{\mathrm{sp},i}$ and $[\eta]_i$ are the specific and intrinsic viscosities, respectively, of the i-th elution volume summation step, ΔV.

In deriving Eqn (7), the effect of band broadening in the SEC column has not been considered. Balke *et al.*[13] have shown that the value of $\overline{M}_n$ obtained from Eqn (7) should be corrected for band broadening. Under acceptable assumptions, the correction factor depends on the standard deviation of the Gaussian spreading function and is independent of the type and breadth of the MWD of the polymer.

SAMPLES AND METHODS

Statistical copolymers of styrene with small amounts of divinylbenzene were prepared by radical polymerization using 2,2′-azobis(2-methylpropanenitrile) as initiator. 1-Octanethiol was employed as a chain-transfer agent to reduce the molecular weight of the primary chains. This enabled the gel point to be shifted to higher conversions. Divinylbenzene was of technical grade, with 80% of mixture of isomers, the residue being mainly 3- and 4-ethylvinylbenzene. At preset intervals of time, samples of the reaction mixture were taken, yielding a series of polymers prepared at increasing degrees of conversion. The conditions of the polymerization reactions are specified in Table 1 and the dependence of the degrees of conversion on time for the three reaction series S1 through S3 can be seen in Fig. 1.

Polydivinylbenzene samples were synthesized (Table 2) by polymerization of a technical mixture of isomers specified above. Polymerization was initiated by 2,2′-azobis(2-methylpropanenitrile) in the presence of the stable nitroxide radical (TEMPO).

The polymers were analysed by SEC measurements with triple light scattering–viscosity–concentration detection: a light-scattering photometer (DAWN DSP-F, Wyatt Technology Corp.) measuring at 18 angles of observation, a modified differential viscometer (Viscotek model TDA 301) and a differential refractometer (Shodex RI 71) were the detectors. The mobile phase was tetrahydrofuran at ambient temperature. As a reference UC, we used that based on measurements of the molecular weight and intrinsic viscosity of 10 linear polystyrene standards

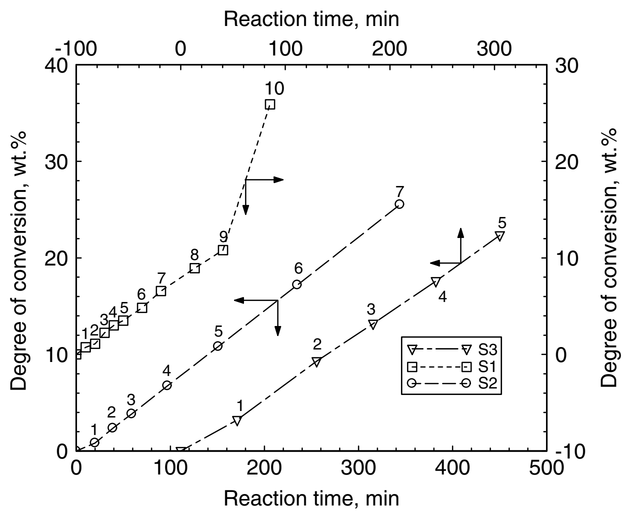

Figure 1. Degrees of conversion *versus* reaction times for copolymers of styrene and divinylbenzene prepared under conditions specified in Table 1.

Table 1. Initial feed of reaction components (mol L^{-1}) for radical statistical copolymerization of styrene and divinylbenzene at 60 °C

Series	ST	DVB	AIBN	OT
S1	5.72	0.026	0.025	0
S2	5.75	0.094	0.031	0.0093
S3	5.66	0.129	0.032	0.0146

Xylene as solvent. ST, styrene; DVB, divinylbenzene; AIBN, 2,2′-azobis(2-methylpropanenitrile); OT, 1-octanethiol.

Polym Int **57**:1152–1158 (2008)
DOI: 10.1002/pi

Table 2. Radical polymerization of divinylbenzene in the presence of TEMPO at 125 °C

Sample	DVB	AIBN	Reaction time (min)	Conversion (mass%)
DV1	5.64	0.02	20	*ca* 20
DV2[a]	2.82	0.02	30	*ca* 10
DV3	5.64	–	60	*ca* 7

DVB and AIBN are the initial concentrations (in mol L^{-1}) of divinylbenzene and 2,2′-azobis(2-methylpropanenitrile), respectively, in the reaction mixture; the initial concentration of nitroxide TEMPO was identical for all the three experiments (0.03 mol L^{-1}).
[a] A dimethylformamide solution of divinylbenzene (1:1 volume ratio) was used.

with narrow MWDs in the range of molecular weights 3.6×10^3–1.6×10^6. The values calculated using Eqn (7) were corrected for band broadening. According to Balke *et al.*,[13] a resolution correction factor that can be applied to the value of $\overline{M}_n$ is given by $\exp(\sigma^2 D^2/2)$, where σ is the constant standard deviation of the Gaussian spreading function and $D = 2.303 \times \mathrm{d}\log M/\mathrm{d}V$ is the slope of the calibration dependence. For our SEC set, $\sigma = 0.25$ and $D = 1.34$. This yields for the resolution correction factor a value of 1.06.

RESULTS AND DISCUSSION

The reference UC plot for linear polystyrene standards is shown in Fig. 2; the plot for all the samples measured, including polystyrene standards, forms a master stripe with a width of ±10% rather than a master curve (Fig. 3). A typical UC dependence of a branched copolymer of styrene and divinylbenzene (sample S1-10) resulting from a SEC measurement with three detectors deviates somewhat from the dependence based on linear standards (Figs 2 and 3). The reason for the aberration is that calibrations or UCs obtained from measurements on a single sample usually have a smaller slope than those based on standards, because the former calibrations are more sensitive to the effect of band broadening.[18] Deviations of UCs for branched polymers from those for linear standards in the region of small elution volumes, where high molecular weight, highly branched fractions elute, may also be caused by the variation of the value of the viscosity function Φ with the type and degree of branching and breadth of the MWD.[6] Poor signal-to-noise ratio at both shoulders of the elution curves, particularly in the region of low molecular weights, is another source of imprecision.[2] This indicates that the Goldwasser approach may be useful for the determination of the number-average molecular weight for polymers with very broad distribution, such as highly branched polymers approaching the gel point, where an error of ±10% may be tolerable. On the other hand, it can hardly be expected to yield satisfactory data for narrow MWD polymers. From the experimental data, the following characteristics of the polymers analysed were calculated:

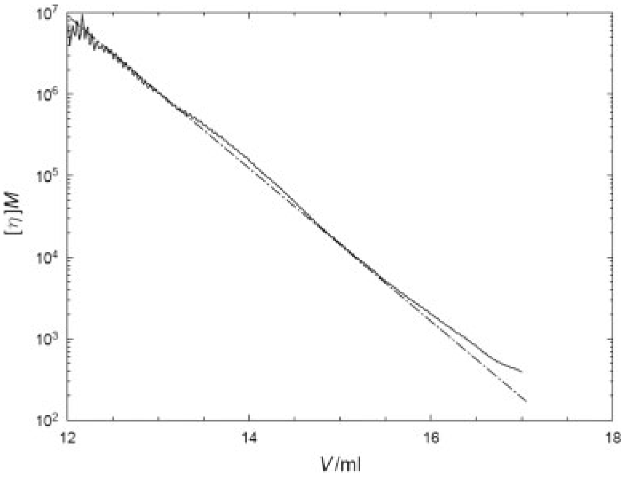

Figure 2. Comparison of local universal calibration for sample S1-10 with that obtained for polystyrene standards (dashed line).

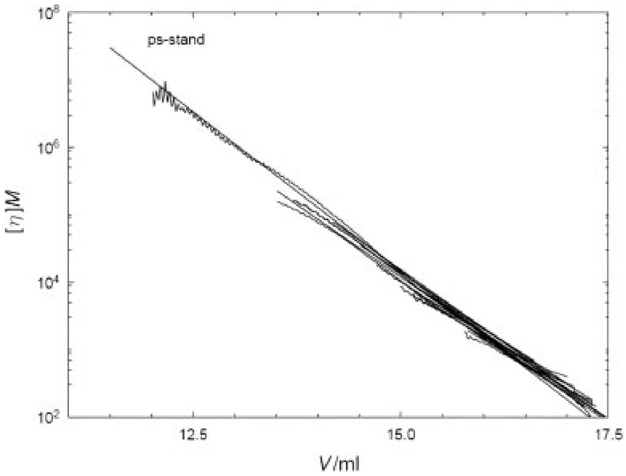

Figure 3. Comparison of local universal calibrations for all branched samples studied with that obtained for polystyrene standards (ps-stand).

- $\overline{M}_{n,LS}$ – number-average molecular weight from the signals of the light-scattering and concentration detectors;
- $\overline{M}_{w,LS}$ – weight-average molecular weight from the signals of the light-scattering and concentration detectors;
- $\overline{M}_{n,PS}$ – polystyrene-equivalent number-average molecular weight from the signal of concentration detector and calibration by linear polystyrene standards;
- $\overline{M}_{w,PS}$ – polystyrene-equivalent weight-average molecular weight from the signal of concentration detector and calibration by linear polystyrene standards;
- $\overline{M}_{n,UC}$ – number-average molecular weight from the signals of the viscosity and concentration detectors, plus UC;
- $\overline{M}_{w,UC}$ – weight-average molecular weight from the signals of the viscosity and concentration detectors, plus UC;

- $\overline{M}_{n,G}$ – number-average molecular weight according to Goldwasser, from the signal of the viscosity detector, the mass of the injected sample and UC;
- $[\eta]$ – intrinsic viscosity from the signals of the viscosity and concentration detectors;
- g' – contraction factor, i.e. ratio of intrinsic viscosities of a branched polymer and a chemically identical linear polymer with the same weight-average molecular weights, $[\eta]_b/[\eta]_l$.

The characteristics of all the polymers studied are collected in Table 3. We have calculated and tabulated the two most important molecular weight averages, $\overline{M}_n$ and $\overline{M}_w$, and their ratios, $\overline{M}_w/\overline{M}_n$, obtained by various methods of processing the experimental data. Data were not smoothed, thus giving an estimate of the accuracy and reproducibility of the measurements.

From the relatively close values of $\overline{M}_{n,UC}$ and $\overline{M}_{n,G}$ in Table 3, it can be concluded that the Goldwasser method is a plausible method for obtaining $\overline{M}_n$ assuming the validity of the UC theorem even for highly branched polymers. One has, however, to keep in mind the fact that the UC itself is associated with an error of some ±10% and this is reflected in the resulting values of $\overline{M}_n$. On the other hand, the independence of the measurements of the refractive index signal makes the Goldwasser method preferable, for example, for the investigation of copolymers with constituents differing in refractive index increments.

Some of the quantities defined above are prone to inherent bias. $\overline{M}_{n,LS}$ is overestimated because the weight-average molecular weights of the slices are measured by light scattering, whereas the number-average molecular weights of the slices should be used to calculate correct $\overline{M}_n$ values for the whole polymer.[3] Moreover, $\overline{M}_n$ is sensitive to the low molecular weight fractions of the sample, for which the signal-to-noise ratio for light-scattering detection is rather poor.[2] Overestimation of $\overline{M}_{n,LS}$ values results in lower than correct values of the ratio $\overline{M}_{w,LS}/\overline{M}_{n,LS}$.

For branched polymers, including branched polystyrene, both $\overline{M}_{n,PS}$ and $\overline{M}_{w,PS}$ values are lower than the correct values because branched molecules with higher molecular weight elute together with linear molecules of lower molecular weight, present even in highly branched samples,[19] and are recorded as linear molecules with the same hydrodynamic volume. A comparison of the values in Table 3 of $\overline{M}_{n,PS}$ with $\overline{M}_{n,UC}$ and $\overline{M}_{w,PS}$ with $\overline{M}_{w,UC}$ shows that both types of number averages are virtually identical for slightly branched polymers and $\overline{M}_{n,PS}$ is perceivably smaller than $\overline{M}_{n,UC}$ only for highly branched polymers. On the other hand, $\overline{M}_{w,PS}$ values are always smaller, and for highly branched polymers much smaller, than the values of $\overline{M}_{w,UC}$. What occurs in SEC separation of branched polymers can be schematically described as a transfer of molecules with high molecular weights and high degrees of branching into the low molecular weight end of the elution. From the definition of the number-average molecular weight, it follows that an admixture of a high molecular weight species does not affect $\overline{M}_n$ much because in a given mass fraction of the transferred polymer the number of added molecules is small. On the contrary, a withdrawal of a portion of highly branched high molecular weight molecules from the high molecular weight end of the elution decreases dramatically the weight-average molecular weight. Thus, both polystyrene-equivalent averages are decreased in SEC measurements of branched polymers but $\overline{M}_{w,PS}$ is affected more than $\overline{M}_{n,PS}$. This results in an underestimation of the value of the ratio $\overline{M}_w/\overline{M}_n$ Users of polystyrene calibration for the characterization of branched polymers should realize that the correct values of the $\overline{M}_w/\overline{M}_n$ ratios may differ from those obtained employing polystyrene calibration by a multiple.

Similarly, one has to keep in mind that both polystyrene-equivalent molecular-weight averages differ from the correct ones if the Mark–Houwink parameters of the polymers measured differ from those for linear polystyrene.[20]

The ratios $\overline{M}_{w,UC}/\overline{M}_{n,UC}$ are typically higher than the values of $\overline{M}_{w,LS}/\overline{M}_{n,LS}$ by some 30%. This is a consequence of systematic overestimation of $\overline{M}_{n,LS}$ values. Probably the ratios $\overline{M}_{w,UC}/\overline{M}_{n,UC}$ are better estimates of the breadth of the MWD than the ratios $\overline{M}_{w,LS}/\overline{M}_{n,LS}$.

Surprisingly enough, the values of $\overline{M}_{w,UC}$ are on average higher than $\overline{M}_{w,LS}$ by 10%. At the present time, we cannot offer any explanation for this finding. The ratio $\overline{M}_{n,G}/\overline{M}_{n,UC}$ varies by ±15% around unity. This roughly corresponds to the breadth of the UC stripe (Fig. 3). The quantity g' is an important and easily accessible characteristic of a branched polymer. If intrinsic viscosity has been measured for a non-uniform branched polymer, the question arises as to what molecular weight average $[\overline{\eta}]_b$ relates. $[\overline{\eta}]_l$ is always a weight-average value and the appropriate, though not rigorous, reference molecular weight is its weight average, $\overline{M}_w$. In SEC measurements of branched polymers, linear molecules elute with branched molecules with higher molecular weights. This results in what is called local non-uniformity[21] at individual elution volumes. Estimates of the extent of local non-uniformity indicate[22] that its effect on the value of $\overline{M}_w$ is not too strong and $\overline{M}_w$ seems to be an adequate reference value.

CONCLUSIONS

1. For statistically branched polymers with high degrees of branching, reasonable estimates of number-average molecular weight are obtained from SEC measurements with an on-line viscosity detector and under the assumption of the validity of the UC theorem.
2. For branched polymers, the UC plot is a stripe of width of 10–15% in $[\eta]M$ values, which results in a corresponding uncertainty in $\overline{M}_n$ values. This degree of uncertainty is acceptable for polymers

Polym Int **57**:1152–1158 (2008)
DOI: 10.1002/pi

Table 3. Number- and weight-average molecular weights (×10^{-3}) of the branched samples, $\overline{M}_n$ and $\overline{M}_w$, obtained by on-line light scattering, calibration with polystyrene standards, universal calibration and the Goldwasser method, denoted by subscripts LS, PS, UC and G, respectively, together with the intrinsic viscosities, [η],and values of g' (see text)

Sample	$\overline{M}_{n,LS}$	$\overline{M}_{w,LS}$	$\frac{\overline{M}_{w,LS}}{\overline{M}_{n,LS}}$	$\overline{M}_{n,PS}$	$\overline{M}_{w,PS}$	$\frac{\overline{M}_{w,PS}}{\overline{M}_{n,PS}}$	$\frac{\overline{M}_{w,PS}}{\overline{M}_{w,LS}}$	$\overline{M}_{n,UC}$	$\overline{M}_{w,UC}$	$\frac{\overline{M}_{w,UC}}{\overline{M}_{n,UC}}$	$\overline{M}_{n,G}$	$\frac{\overline{M}_{n,LS}}{\overline{M}_{n,UC}}$	$\frac{\overline{M}_{n,LS}}{\overline{M}_{n,UC}}$	$\frac{\overline{M}_{n,G}}{\overline{M}_{n,LS}}$	$\frac{\overline{M}_{n,G}}{\overline{M}_{n,UC}}$	[η] (mL g^{-1})	g'
S1-1	44.9	72.2	1.61	35.6	66.6	1.87	0.92	37.1	69.3	1.87	41.0	1.21	1.04	0.91	1.11	0.371	0.925
S1-2	49.0	69.2	1.41	39.9	68.4	1.71	0.99	38.8	72.2	1.86	39.0	1.26	0.96	0.80	1.01	0.368	0.946
S1-3	49.6	70.9	1.43	41.2	70.8	1.72	1.00	41.0	75.6	1.84	47.0	1.21	0.94	0.95	1.15	0.372	0.939
S1-4	52.7	73.7	1.40	42.2	74.5	1.77	1.01	42.2	80.3	1.90	45.8	1.25	0.92	0.87	1.09	0.380	0.934
S1-5	51.0	73.1	1.43	41.8	75.2	1.80	1.03	41.8	82.0	1.96	43.2	1.22	0.89	0.85	1.03	0.384	0.948
S1-6	53.4	76.3	1.43	44.2	77.1	1.74	1.01	43.8	84.5	1.93	39.8	1.22	0.90	0.75	0.91	0.383	0.918
S1-7	52.0	77.8	1.50	43.7	79.7	1.82	1.02	42.5	88.4	2.08	35.2	1.22	0.88	0.68	0.83	0.393	0.928
S1-8	55.0	84.8	1.54	46.7	85.6	1.83	1.01	45.2	95.3	2.11	38.3	1.22	0.89	0.70	0.85	0.400	0.888
S1-9	53.7	87.9	1.64	44.2	84.6	1.91	0.96	43.6	94.1	2.16	49.8	1.23	0.93	0.93	1.14	0.396	0.859
S1-10	49.5	114.4	2.31	40.4	93.8	2.32	0.82	41.6	120.2	2.89	49.3	1.19	0.95	1.00	1.19	0.408	0.731
S2-1	6.7	8.0	1.19	5.2	7.8	1.50	0.97	5.1	8.3	1.63	6.1	1.31	0.96	0.91	1.20	0.088	1.052
S2-2	6.1	8.9	1.46	5.4	8.7	1.61	0.98	5.5	9.2	1.67	5.9	1.11	0.97	0.97	1.07	0.094	1.044
S2-3	7.2	10.1	1.40	5.8	9.9	1.71	0.98	5.7	10.7	1.88	6.4	1.26	0.94	0.89	1.12	0.100	1.019
S2-4[a]	–	13.6	1.28	6.5	12.6	1.94	0.93	6.4	13.9	2.17	7.6	–	0.98	–	1.19	0.114	0.936
S2-5	9.9	19.7	1.99	8.0	18.7	2.34	0.95	7.7	21.0	2.73	8.3	1.29	0.94	0.84	1.08	0.144	0.904
S2-6	12.2	38.5	3.16	10.6	35.5	3.35	0.92	10.5	44.1	4.20	9.1	1.16	0.87	0.75	0.87	0.197	0.770
S2-7	20.3	137.0	6.75	20.3	114.0	5.62	0.83	20.3	169.0	8.33	22.4	1.00	0.81	1.10	1.10	0.330	0.520
S3-1	5.4	7.4	1.37	5.7	7.7	1.35	1.04	5.5	8.3	1.51	6.4	0.98	0.89	1.19	1.16	0.086	1.083
S3-2	7.0	12.8	1.83	7.5	12.8	1.71	1.00	6.9	14.6	2.12	7.0	1.01	0.88	1.00	1.01	0.113	0.972
S3-3	8.2	18.8	2.29	9.0	19.9	2.21	1.06	8.7	23.1	2.66	8.6	0.94	0.81	1.05	0.99	0.143	0.930
S3-4	9.8	33.8	3.45	11.0	32.8	2.98	0.97	9.9	34.7	3.51	9.3	0.99	0.97	0.95	0.94	0.172	0.737
S3-5	12.3	71.9	5.85	13.7	63.8	4.66	0.89	14.7	93.4	6.35	16.5	0.84	0.77	1.34	1.12	0.232	0.580
DV-1	21.3	401.6	18.85	13.4	121.3	9.05	0.30	17.5	337.0	19.26	21.3	1.22	1.19	1.00	1.22	0.152	0.111
DV-2	5.9	8.2	1.39	4.3	5.3	1.23	0.65	5.3	7.6	1.43	5.9	1.11	1.08	1.00	1.11	0.053	0.620
DV-3	11.3	25.6	2.27	7.8	15.1	1.94	0.59	10.2	29.0	2.84	12.0	1.11	0.88	1.06	1.18	0.079	0.413

[a] $\overline{M}_{n,LS}$ was not evaluated because of strong noise on the light-scattering signal on the low molecular weight shoulder.

with broad MWD, such as that occurring with statistically highly branched polymers.

3. $\overline{M}_n$ and $\overline{M}_w$ values and their ratios calculated by various methods for the evaluation of the data from SEC with on-line concentration, light-scattering and viscosity detection have been analysed and the reasons for their occasional divergence discussed.
4. Usage of polystyrene calibration in the characterization of branched polymers results in an underestimation of the breadth of MWDs as described by the ratio $\overline{M}_w/\overline{M}_n$.

ACKNOWLEDGEMENTS

The authors acknowledge the support of the Academy of Sciences of the Czech Republic (projects AVOZ 40500505, A4050403 and A400500703) and express their gratitude to Wyatt Technology Ltd for kindly providing the DAWN laser photometer.

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